Economic History II

The Reformation: causes and expropriation

FGV EPGE

Outline of the class

- The Reformation in two classes: the **causes** today, the consequences from here on.
- Why it was not crushed, as earlier reform movements had been: Iyigun (2008).
- Who took the Church's land once the Reformation won, and what they did with it: Cantoni, Dittmar and Yuchtman (2018).

The setting

- October 1517: Martin Luther circulates his 95 Theses, criticizing the price of salvation and the sale of indulgences.
- On the eve of the Reformation the Catholic Church held a **virtual monopoly** in the market for religion in Western Europe.
- Rich accordingly: tithes, sacramental fees, vast land holdings. The foundation stone of St. Peter's Basilica was laid in 1506.
- The Holy Roman Empire: an imperial federation of hundreds of semiautonomous principalities, ecclesiastical states and Free Imperial cities.

Timeline

- **1517**: Luther circulates the 95 Theses from Wittenberg.
- **1521**: the Edict of Worms condemns him as a heretic.
- **1530**: the Schmalkaldic League, a military alliance of Protestant princes.
- **1546-1547**: the Schmalkaldic War.
- **1555**: the Peace of Augsburg. Each prince chooses the religion of his territory (*cuius regio, eius religio*).
- **1618-1648**: the Thirty Years' War.

Why economists keep coming back to this

- Weber (1904): the **calling**, labor sanctified as a task set by God, wealth accumulation morally approved.
- Durkheim (1897): Protestant religious individualism, weaker communal ties, higher suicide rates.
- Marx (1844): religion as "opium of the people", needed to alleviate the misery of poor economic circumstances.
- Becker and Woessmann (2009): not the ethic but human capital. To read the Bible you first learn to read.
- Today's papers come before all of this. One asks why the Reformation survived, the other who got the Church's wealth.

Table 5: Socioeconomic Causes of the Reformation, of its Spread, and of Protestant Religiosity

Study	Setting	Method	Summary
Causes of adoption and spread of Reformation			
Ekelund, Hébert, and Tollison (2002)	Reformation	Theory, narrative	Monopolistic practices of medieval Catholic Church encouraged Protestant entry in entrepreneurial societies with declining feudalism and unstable wealth distribution
Kim and Pfaff (2012)	Germany, 1517-1545	Empirical	Cities with students in Protestant rather than Catholic universities before 1523 more likely to enact Reformation
Becker, Hsiao, Pfaff, and Rubin (2020)	Germany before 1530	Empirical	Cities with personal ties (letters, visits, Wittenberg students) to Luther before 1523 more likely to enact Reformation; interacts with spatial diffusion along trade routes
Cantoni (2012)	Germany, 1600	Empirical	Distance to Wittenberg, ecclesiastical rule, and political and economic power reduce Protestantism
Curuk and Smulders (2019)	Germany, 1600	Empirical	Agricultural potential and smaller population 1500 increase Protestantism
Rubin (2014)	European cities, 1500-1600	Empirical	Having printing press by 1500 increases Protestant adoption by 1600
Boerner, Rubin, and Severgnini (2018)	European cities, 1450-1600	Empirical	Indirect effect of mechanical clock on Reformation through printing press
Dittmar and Seabold (2020)	Germany, 1454-1600	Empirical	Printing competition increased Protestant book content and Reformation adoption
Iyigun (2008)	Europe, 1521-1650	Empirical	Ottoman incursions reduced Catholic-Protestant conflicts within Europe
Causes of Protestant secularization			
Becker and Woessmann (2013)	Prussia, 1886-1911	Empirical	No effect of income on church attendance in panel models
Becker, Nagler, and Woessmann (2017)	Germany, 1890-1930	Empirical	Advanced-school expansion reduced church attendance in panel models

Why did this reform survive?

Iyigun (2008), *Quarterly Journal of Economics*

Luther and Suleyman

LUTHER AND SULEYMAN*

MURAT IYIGUN

Various historical accounts have suggested that the Ottomans' rise helped the Protestant Reformation as well as its offshoots, such as Zwinglianism, Anabaptism, and Calvinism, survive their infancy and mature. Utilizing a comprehensive data set on violent confrontations for the interval between 1401 and 1700 CE, I show that the incidence of military engagements between the Protestant Reformers and the Counter-Reformation forces between the 1520s and 1650s depended negatively on the Ottomans' military activities in Europe. Furthermore, I document that the impact of the Ottomans on Europe went beyond suppressing ecclesiastical conflicts only: at the turn of the sixteenth century, Ottoman conquests lowered the number of all newly initiated conflicts among the Europeans roughly by 25 percent, while they dampened all longer-running feuds by more than 15 percent. The Ottomans' military activities influenced the length of intra-European feuds too, with each Ottoman-European military engagement shortening the duration of intra-European conflicts by more than 50 percent. Thus, while the Protestant Reformation might have benefited from—and perhaps even capitalized on—the Ottoman advances in Europe, the latter seems to have played some role in reducing conflicts within Europe more generally.

"Modern history of Europe begins under stress of the Ottoman conquest."
Lord Acton (1834–1902).

"Mamma, li Turchi!"
Anonymous, Italy.

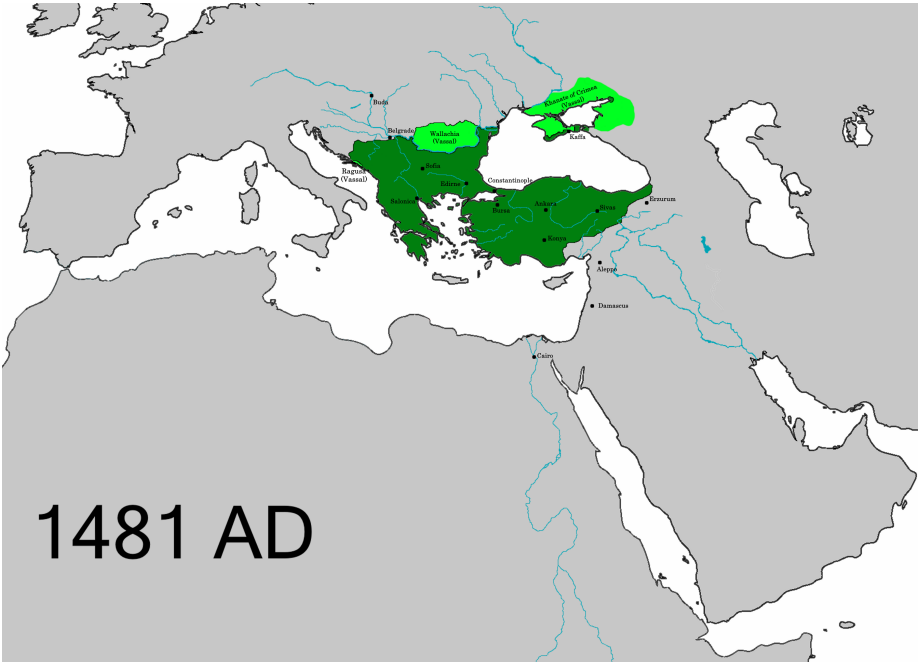
The puzzle

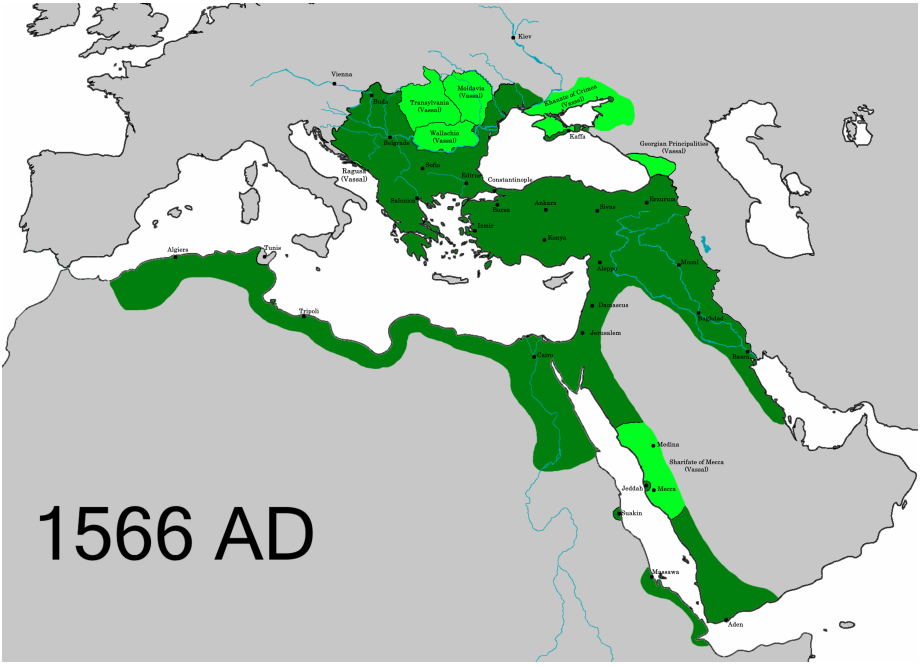
- Luther was not the first challenge to the ecclesiastical monopoly, only the first that survived.
- Wycliffe was fired from Oxford and died in 1384; the movement he inspired, Lollardy, was suppressed and had to go underground.
- Jan Hus was burned alive in Constance in 1415.
- The Albigensian-Cathar Crusade was launched by Pope Innocent III in 1209 and lasted twenty years.
- Lord Acton: "Modern history of Europe begins under stress of the Ottoman conquest."

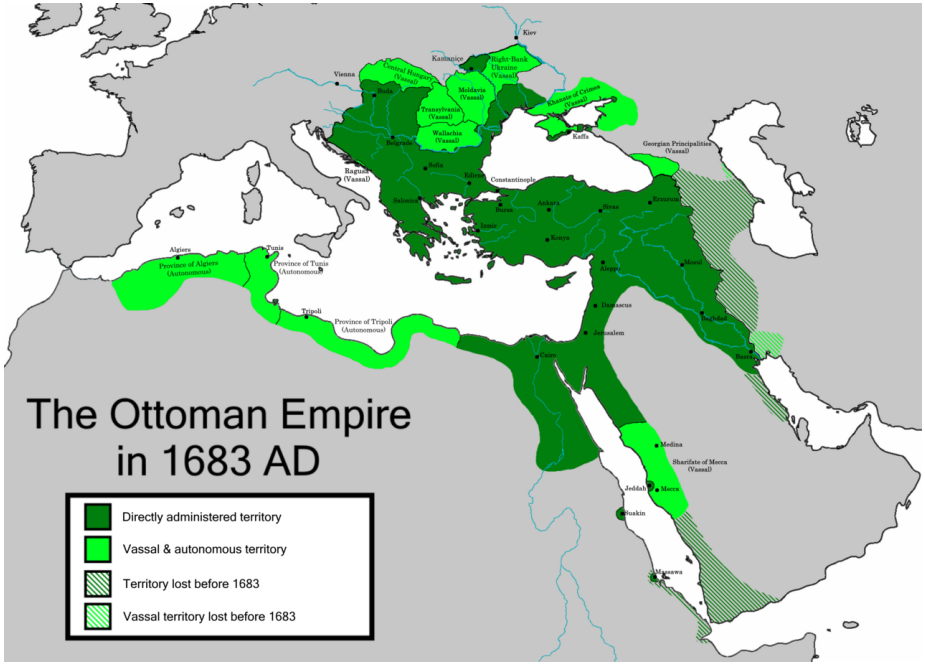
The Ottoman rise

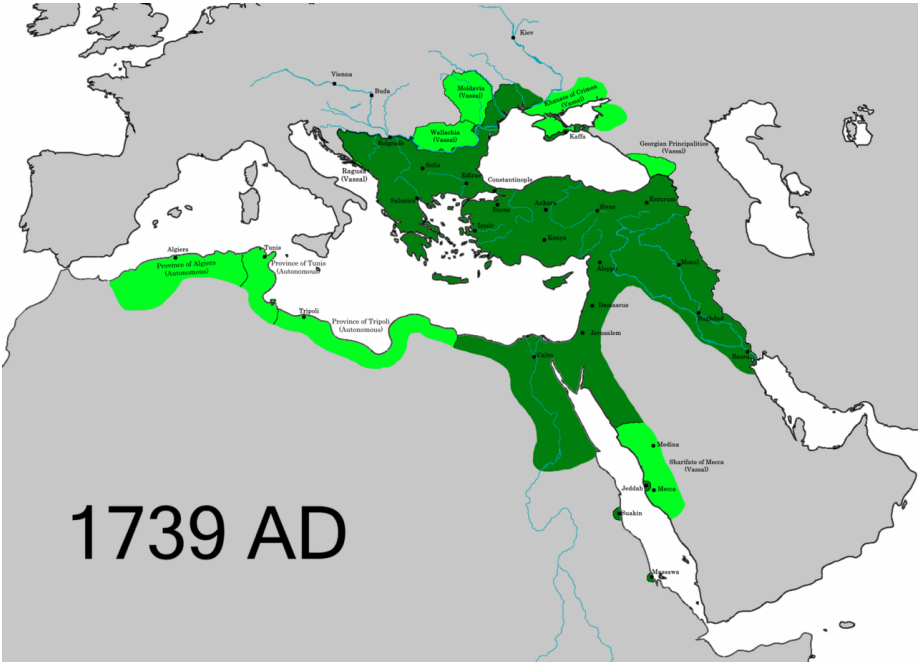
- Constantinople taken in 1453, ending the East Roman Empire. Standard dating of Ottoman dominance: 1453 to the Treaty of Karlowitz in 1699.
- A garrison on the Italian peninsula, at Otranto, in 1481. Belgrade in 1521. Hungary broken at Mohacs in 1526.
- Vienna, the Hapsburg capital, under siege in 1529.
- The empire looked steadily **westward**: early sultans saw themselves as heirs to the eastern Roman Empire.
- Peak under Suleyman I, the Magnificent, 1521 to 1566. Exactly the Reformation's first generation.




Buda
Belgrade
Wallachia (Vassal)
Khanate of Crimea (Vassal)
Kaffa
Ragusa (Vassal)
Sofia
Edirne
Constantinople
Bursa
Ankara
Sivas
Erzurum
Konya
Aleppo
Damascus
Cairo
1481 AD


Kiev
Vienna
Moldavia (Vassal)
Transylvania (Vassal)
Wallachia (Vassal)
Khanate of Crimea (Vassal)
Kaffa
Georgian Principalities (Vassal)
Ragusa (Vassal)
Sofia
Constantinople
Bursa
Ankara
Sivas
Erzurum
Izmir
Konya
Algiers
Tunis
Mosul
Aleppo
Tripoli
Damascus
Jerusalem
Baghdad
Basra
Cairo
Medina
Sharifate of Mecca (Vassal)
Jeddah
Mecca
Suakin
Massawa
Aden
1566 AD


The Ottoman Empire
in 1683 AD
Directly administered territory
Vassal & autonomous territory
Territory lost before 1683
Vassal territory lost before 1683
Kiev
Kaniapico
Right-Bank Ukraine (Vassal)
Vienna
Buda
Central Hungary (Vassal)
Moldavia (Vassal)
Transylvania (Vassal)
Wallachia (Vassal)
Khanate of Crimea (Vassal)
Kaffa
Georgian Principalities (Vassal)
Ragusa (Vassal)
Sofia
Edirne
Constantinopolis
Salonica
Ankara
Bursa
Brusa
Izmir
Konya
Algiers
Province of Algiers (Autonomous)
Tunis
Province of Tunis (Autonomous)
Tripoli
Province of Tripoli (Autonomous)
Aleppo
Mosul
Damascus
Jerusalem
Medina
Sharifate of Mecca (Vassal)
Jeddah
Mecca
Suakin
Massawa


Kiev
Vienna
Buda
Moldavia (Vassal)
Khanate of Crimea (Vassal)
Wallachia (Vassal)
Kaffa
Georgian Principalities (Vassal)
Ragusa (Vassal)
Sofia
Edirne
Constantinople
Bursa
Ankara
Izmir
Konya
Algiers
Province of Algiers (Autonomous)
Tunis
Province of Tunis (Autonomous)
Tripoli
Province of Tripoli (Autonomous)
Damascus
Jerusalem
Medina
Sharifate of Mecca (Vassal)
Jeddah
Mecca
Masawn
1739 AD

Reading the four maps

- 1481 to 1566 is the expansion that matters here: the Balkans, Hungary, the approaches to Vienna.
- After 1566 the frontier moves very little, then recedes. The threat has a peak and a decline.
- Ottoman expansion also consolidated Europe politically: bigger states to sustain bigger military establishments.
- The fear was concrete. In Italy children were frightened with "*Mamma, li Turchi!*" (Mummy, the Turks!)

The Protestants' bargaining chip

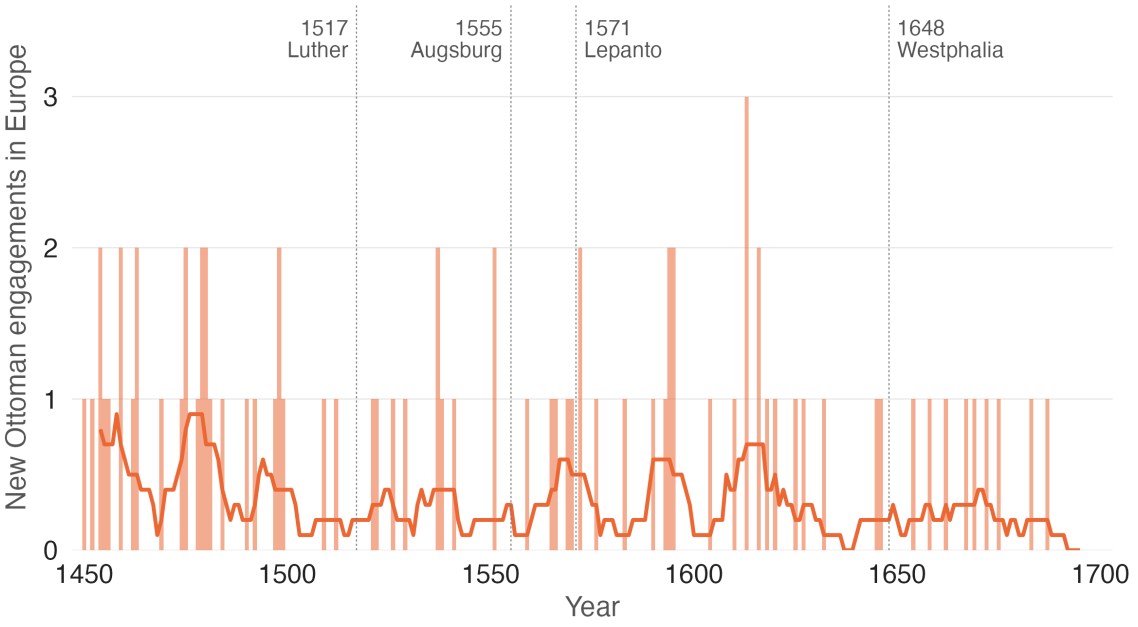
- Two effects at once: urgency for the pope, Charles V and his brother Ferdinand, and **leverage** for a movement in its infancy.
- Ferdinand needed manpower and money against the Turks. Protestant leaders tied every commitment to concessions on religion.
- Inalcik: "at first Luther and his adherents followed a passive course, maintaining that the Ottoman threat was a punishment from God". Then they helped, at a price.
- Shaw, on Suleyman's advance: "causing Charles to make concessions to the Protestants in Germany to gain their support, a major factor in the subsequent survival and expansion of the Lutheran movement".
- Treaty of Passau, then the Peace of Augsburg in 1555: the Empire officially recognizes the Lutheran movement's right to exist.

The data

- Brecke's *Conflict Catalog*: every recorded violent conflict worldwide from 1400 CE on, virtually complete for Europe, North Africa and the Near East.
- Threshold: Richardson magnitude 1.5, that is 32 or more deaths. Lethal violence between organized groups with conflicting political goals; the state need not be an actor.
- **251 annual observations**, 1450 to 1700, chosen to match the Ottoman era of dominance.
- Population from McEvedy and Jones; distances from a city distance tool, measured from Istanbul.

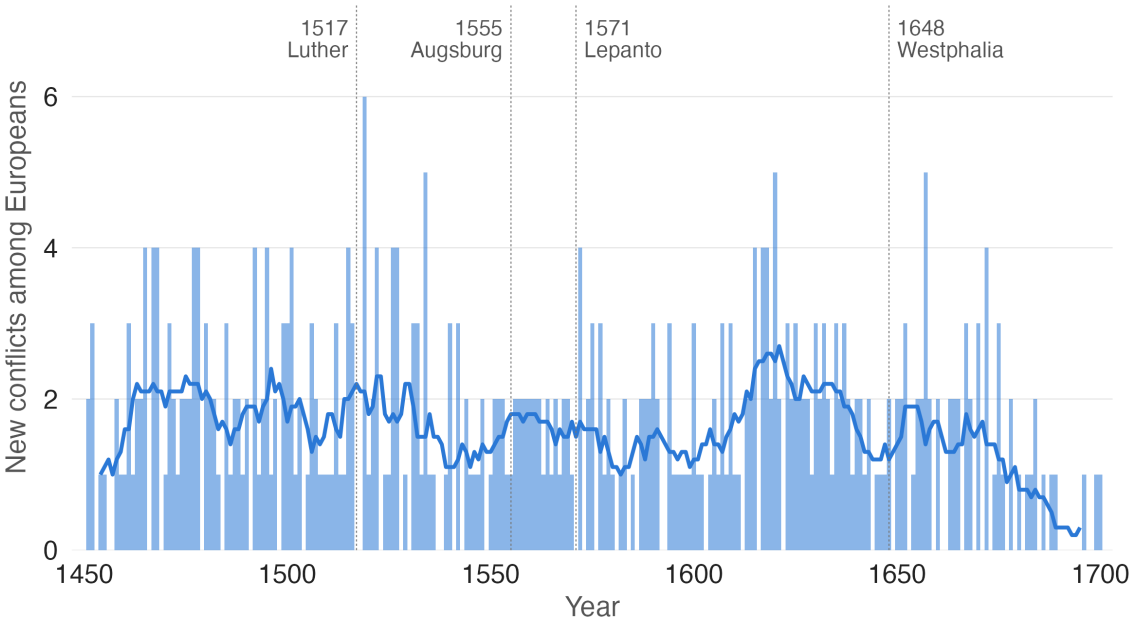
What is counted each year

- Ottoman military engagements against European powers, splitting Eurasia vertically at Istanbul.
- Separately: Ottoman fighting elsewhere, including their own domestic discords. This should push the **other way**.
- Three outcomes: conflicts newly started within continental Europe; all conflicts running, new or old; and Catholic-Protestant confrontations.
- Why count running conflicts too: preindustrial warfare was seasonal, halting in winter and resuming in spring, so every unresolved war was renewed each year.


1517
Luther
1555
Augsburg
1571
Lepanto
1648
Westphalia
New Ottoman engagements in Europe
0
1
2
3
1450
1500
1550
1600
1650
1700
Year

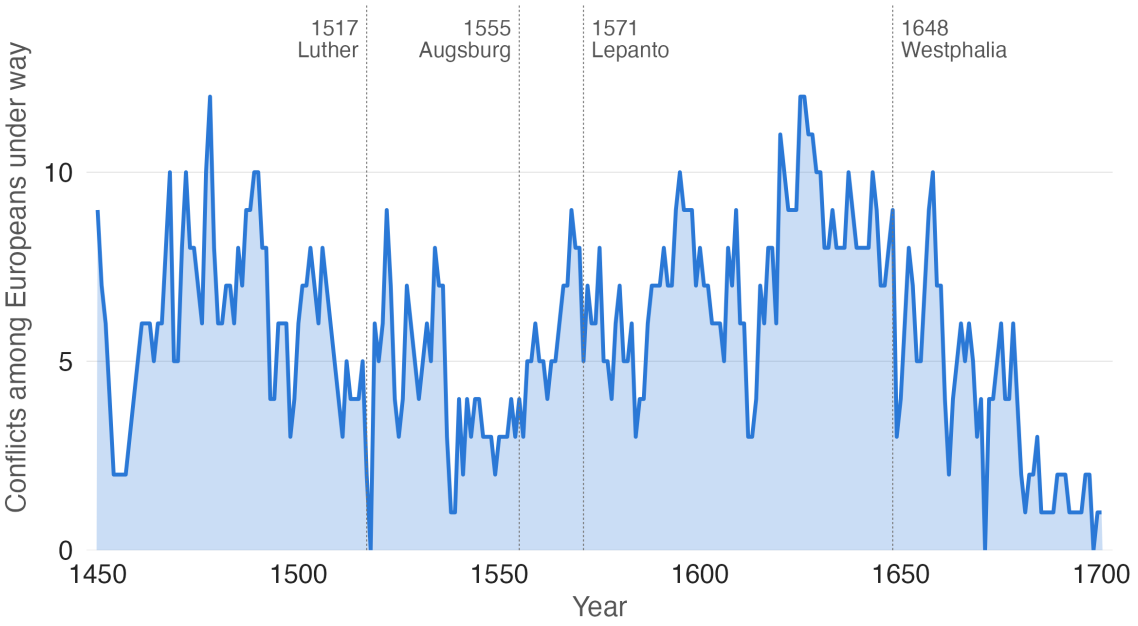
Ottoman engagements in Europe, 1450 to 1700

- Each bar: new Ottoman campaigns or battles against European powers that began that year. Rebuilt from Brecke's catalog with the paper's definitions.
- About one new engagement every three years, in **waves**: 1450s to 1480s, 1520s to 1540s, 1560s to 1570s, 1590s to 1620s.
- Quiet stretches around 1500 to 1515, in the 1580s and in the 1640s.
- The line smooths the bars over ten years so the waves stand out.


1517
Luther
1555
Augsburg
1571
Lepanto
1648
Westphalia
New conflicts among Europeans
0
2
4
6
1450
1500
1550
1600
1650
1700
Year

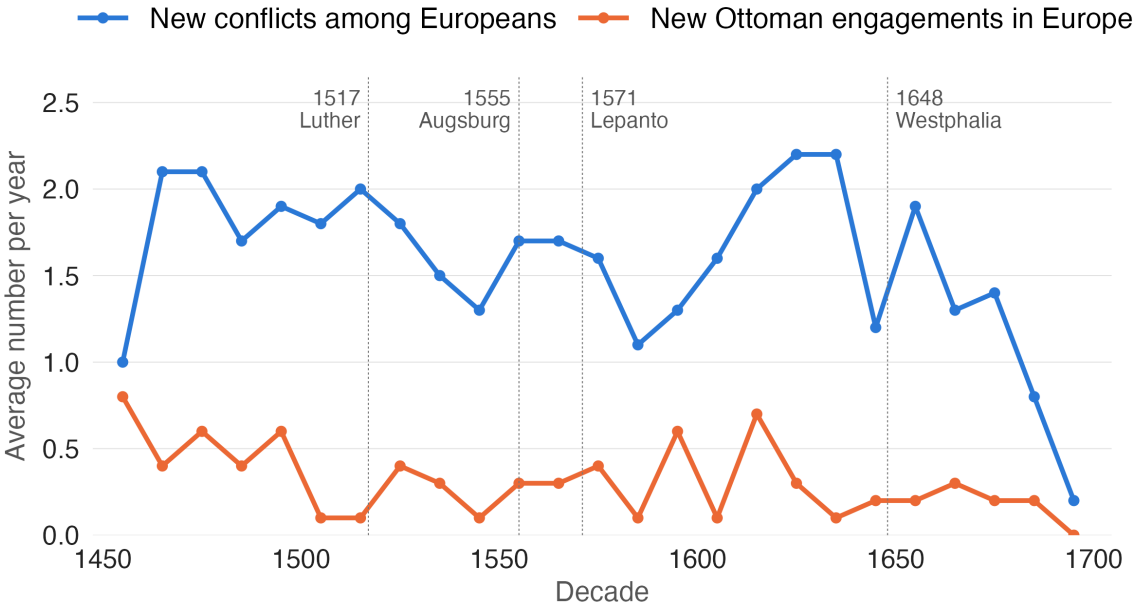
New conflicts among Europeans, 1450 to 1700

- Each bar: violent conflicts that began that year among or within continental European states. Britain and Scandinavia left out unless a continental power took part.
- Roughly **1.5 new conflicts a year**, the figure the paper reports.
- The 1540s to 1550s and the 1580s are unusually calm. The 1620s to 1630s are the busiest years.


1517
Luther
1555
Augsburg
1571
Lepanto
1648
Westphalia
Conflicts among Europeans under way
0
5
10
1450
1500
1550
1600
1650
1700
Year

Conflicts among Europeans under way each year

- Here a war counts in every year it lasts, not only in the year it starts.
- Typically **five to eight wars** under way at once. The Thirty Years' War pushes the count above ten.
- The paper asks whether Ottoman campaigns also shortened wars already under way, not only delayed new ones.
- The catalog thins out after 1690, so the drop at the very end is partly missing records.


New conflicts among Europeans
New Ottoman engagements in Europe
1517
Luther
1555
Augsburg
1571
Lepanto
1648
Westphalia
2.5
2.0
1.5
1.0
0.5
0.0
Average number per year
1450
1500
1550
1600
1650
1700
Decade

The two series side by side

- Both series averaged by decade, so the eye can compare the waves.
- Some episodes fit the story: Ottoman pushes in the 1520s to 1530s and 1590s to 1610s come with **dips** in new European wars, and the quiet Ottoman 1500s come with a busy Europe.
- Others do not: in the 1450s to 1470s both are high at once.
- On the raw picture the two do not move in opposite directions overall. The paper's claim comes from comparing year by year, holding fixed the long decline in warfare and the previous year's wars.

Result: all European conflict

- Baseline rates, 1451 to 1700: about 1.5 conflicts newly started per year among Europeans, about 4.7 running in total.
- One more Ottoman engagement in Europe around 1500 lowers newly initiated intra-European conflicts by at least **25 percent**.
- Conflicts already running: roughly 20 to 25 percent fewer around the turn of the sixteenth century.
- Ottoman fighting outside Europe shows no comparable effect on European conflict.
- The effect does not weaken with distance from the frontier. Deep western Europe is affected too.

Result: wars of religion

- Religiously motivated conflicts inside Europe averaged 0.360 per year over 1451 to 1700.
- An extra Ottoman excursion around 1500 dampens religious strife by something on the order of 30 percent.
- Narrow to Catholic-Protestant confrontations only, 1521 to 1650: about 0.615 running feuds per year.
- One Ottoman engagement with a European foe cuts that by roughly **40 to 50 percent** around the turn of the sixteenth century.
- One catch: Catholic-Protestant wars cannot exist before 1517 and stop with Westphalia in 1648. So this result rests on about 130 years of data, half the sample, and is less precise than the others.

Result: length and severity

- An average intra-European conflict, 1451 to 1700, lasted about 2.4 years.
- Ottoman engagements shortened that by about **60 percent**. Ottoman fighting elsewhere lengthened European feuds.
- Severity, from Levy's independent data: battle deaths per million European population averaged 723.
- A drop of about 110, roughly 15 percent, from Ottoman engagements. Different source, same sign.

Could the causation run backwards?

- You would need a reason for the Ottomans to attack precisely when Europe was at peace with itself.
- The plausible reverse story goes the other way: strike when Europe is divided. That **attenuates** the estimates.
- Using last year's Ottoman activity instead of this year's: same results, often stronger.
- Where sultans campaigned was partly personal: the age at accession varied wildly (Mehmed II was 12, Selim II was 42 when his father Suleyman died), and harem factions lobbied over the geography of conquest.

When the effect stops

- The suppressing effect wanes over the sample and turns positive around 1593 in the baseline specification.
- Roughly two decades after **Lepanto**, 1571: the Holy Roman Empire fleet decimated the entire Ottoman navy.
- Lepanto was the first major European victory over the Turks, and the eastern Mediterranean never returned to Ottoman dominance.
- Counter-Reformation in earnest after 1571. The Thirty Years' War, 1618 to 1648, the most murderous sectarian phase, falls in a period of Ottoman weakness.
- By then the German Protestants were no longer a budding movement but a formidable opponent.



Europeans who dealt with the Ottomans

- Popes Innocent VIII and Alexander VI cooperated with Sultan Bayezit for nonaggression and a subsidy.
- Francis I signed a treaty with Suleyman in 1535: French trade throughout the empire at Turkish rates, and a protectorate over Catholics in the Levant.
- But outright military cooperation is rare: three instances, none before 1551. Under 10 percent of Ottoman European campaigns from 1550, 3 percent over 1401 to 1700.
- It also carried a **stigma**: Francis and Charles V settled under pressure from European courts and from a pope who wanted Europe united against Islam.



The net toll

- Two opposing effects on European deaths: fewer and shorter wars among Europeans, more deaths in Ottoman-European fighting itself.
- On balance, Ottoman conflicts account for somewhere between a 2 and 17 percent **drop** in total European deaths.
- Over the narrower 1451 to 1650 window, roughly 20 percent.
- Indirect alliances are recorded in the data as an Ottoman-European conflict plus an intra-European one, which biases against the finding.

Summing up Iyigun

- An external threat suppressed European conflict in general, and Catholic-Protestant conflict inside that.
- So Protestant survival may be partly a **by-product** of a Europe too busy to fight itself, not a special property of Protestantism.
- The schisms themselves can be taken as given: the record of splits predates the Ottomans by a millennium (Assyrian 431, Coptic 451, the Great Schism of 1054).
- Protestantism survived because of what was happening on the Hungarian frontier, not because of anything in its doctrine.

Table 4: Socioeconomic Consequences of the Reformation

Study	Setting	Method	Summary
Protestantism, human capital, and socioeconomic outcomes			
Becker and Woessmann (2009)	Prussia, late 19th c	Empirical	Human capital theory of Protestant economic history: Protestantism increased literacy and economic outcomes
Becker and Woessmann (2010)	Prussia, 1816	Empirical	Effect of Protestantism on education already before industrialization
Becker and Woessmann (2008)	Prussia, 19th c	Empirical	Protestantism decreased gender gap in education
Boppart et al. (2013, 2014)	Switzerland, late 19th c	Empirical	Effect of Protestantism on educational spending and performance in Switzerland
Becker, Cinnirella, and Woessmann (2010)	Prussia, 1849-1905	Empirical	Protestantism-induced education reduced fertility
Becker, Hornung, and Woessmann (2011)	Prussia, 19th c	Empirical	Protestantism-induced education increased non-textile industrialization in both phases of Industrial Revolution
Hornung (2014)	Brandenburg, 1700, 1802	Empirical	Long-run effect of Huguenot immigrants on productivity of textile manufactories
Reformation, Protestant ethic, and socioeconomic outcomes			
Cantoni (2015)	Germany, 1300-1900	Empirical	No effect of Protestantism on German city growth
Dittmar and Meisenzahl (2020)	Germany, 1520-1819	Empirical	Adoption of church ordinances after plagues increased upper-tail human capital and city population growth
Spenkuch (2017)	Germany, 17th c, today	Empirical	Protestant areas in 1624 have longer working hours, but not hourly wages, today
Basten and Betz (2013)	Switzerland, 16th c, today	Empirical	Protestantism reduced political preferences for leisure time, redistribution, and government intervention
Beatton, Skali and Torgler (2019)	Germany, WWII	Empirical	Protestant soldiers expend more effort than Catholic soldiers in WWII
Blum and Strebel (2016)	Germany, WWI	Empirical	Height decrease of cohort born in WWI lower for Protestants than Catholics
Cantoni, Dittmar, and Yuchtman (2018)	Germany, 1480-1600	Empirical	Reallocation of human and physical capital investment from religious to secular purposes in immediate aftermath of Reformation
Heldring, Robinson, and Vollmer (2017)	England, 16th-18th c	Empirical	Dissolution of monasteries increased agricultural and industrial productivity
Negative socioeconomic consequences of Reformation			
Becker and Woessmann (2018)	Prussia, early and late 19th c	Empirical	Protestantism increased suicide; social cohesion as mechanism
Leeson and Russ (2018)	Europe, 1500-1700	Empirical	Catholic-Protestant battles increased witch trial incidence

What did the Reformation move?

Cantoni, Dittmar and Yuchtman (2018), *Quarterly Journal of Economics*

Religious competition and reallocation

Religious Competition and Reallocation: The Political Economy of Secularization in the Protestant Reformation

Davide Cantoni, Jeremiah Dittmar, and Noam Yuchtman

NBER Working Paper No. 23934
October 2017
JEL No. E02,J24,N13,N33

ABSTRACT

Using novel microdata, we document an unintended, first-order consequence of the Protestant Reformation: a massive reallocation of resources from religious to secular purposes. To understand this process, we propose a conceptual framework in which the introduction of religious competition shifts political markets where religious authorities provide legitimacy to rulers in exchange for control over resources. Consistent with our framework, religious competition changed the balance of power between secular and religious elites: secular authorities acquired enormous amounts of wealth from monasteries closed during the Reformation, particularly in Protestant regions. This transfer of resources had important consequences. First, it shifted the allocation of upper-tail human capital. Graduates of Protestant universities increasingly took secular, especially administrative, occupations. Protestant university students increasingly studied secular subjects, especially degrees that prepared students for public sector jobs, rather than church sector-specific theology. Second, it affected the sectoral composition of fixed investment. Particularly in Protestant regions, new construction shifted from religious toward secular purposes, especially the building of palaces and administrative buildings, which reflected the increased wealth and power of secular lords. Reallocation was not driven by preexisting economic or cultural differences. Our findings indicate that the Reformation played an important causal role in the secularization of the West.

The question

- How does religious competition change the allocation of resources between religious and secular uses?
- The usual economics of religion has one market: **salvation**, producers and believers.
- This paper adds the market the historians care about: state authorities buying political legitimacy from religious elites.
- Entry by a competitor moves both markets at once, and the second one is where the resources are.
- Setting: the German-speaking lands of the Holy Roman Empire, 1480 to 1600.

The old equilibrium

- A monopolist producer selling salvation to believers and legitimacy to rulers, both at a high price.
- The ruler's price was paid in control over resources, tax exemptions, and some political power.
- The Church controlled roughly **one-third** of all farmland in the principality of Wuŕttemberg, and about 26 percent in the lordship of Ruppin.
- Salvation was expensive by design: services in Latin, a specialist clergy, seven sacraments, indulgences sold against time in purgatory.

Bargaining was already old

- Investiture Controversy: Gregory VII excommunicated Henry IV, withdrawing his legitimacy. Henry begged for readmission, barefoot in the snow at Canossa, in 1077.
- Concordat of Worms, 1122: secular lords got de facto power over the appointment of bishops, the Church kept the de jure right.
- European princes claimed authority **directly from God**, so they leaned on religious endorsement less than rulers elsewhere.
- Charles V, the Church's great defender, was himself taking Church assets just before the Reformation, among them the domain of the bishop of Utrecht.

What the entrant offered

- In the salvation market: the believer's direct relationship with God, salvation by faith alone, sacraments cut from seven to two, far less clergy needed.
- Cameron: "For laypeople, the point about the Reformation was that it abolished the expensive and complicated apparatus to which they had resorted so regularly for the good of their souls."
- In the legitimacy market: reformers needed the lords, so they sold legitimacy **cheap**, conceding extensive control of church resources.
- Catholics could not match the offer. Changing the price would have meant changing doctrine.

The attack on monasticism

- Monastery wealth was the largest share of the Catholic Church's income-producing real estate, and it was popular to attack.
- A declaration from the 1520s: "everywhere there are too many monasteries, and that they unashamedly claim to be outside the world", so "we have considered together and decided to tolerate no monastery any longer, but to close them".
- Expropriation was tempting but needed a **legal** justification. Protestant theologians supplied it in memoranda "constructed around meticulous citations of Canon and Roman law".
- Two arguments: the lord was preventing misappropriation by Catholic agents, and acting as the legally recognized lay administrator of church property.

The fight over title

- Lords began expropriating in the 1520s; the Church sued for restitution at the Reichskammergericht, the Empire's highest court.
- After a purge of jurists there, the **Schmalkaldic League** formed in 1530, with keeping former church property among its objectives.
- Its official 1540 position paper addresses property rights first and religious doctrine second.
- Some rulers moved cautiously: the elector of Saxony only put confiscated property under his own chancellery in the 1540s.
- Title was effectively secured only by the settlement after the Schmalkaldic War of 1545-1546.

Data: monasteries

- **3,094 monasteries** from Jürgensmeier and Schwerdtfeger: location, foundation date, closure date where applicable.
- Foundation dates recovered for 3,085 of the 3,094, using territorial archives and the *Deutsches Städtebuch* where the catalog is silent.
- Over 2,200 towns; for each town, the number of monasteries open within 25 km in every year.
- Each town assigned to its secular lord as of 1500, and that lord's religion coded from Cantoni (2012).

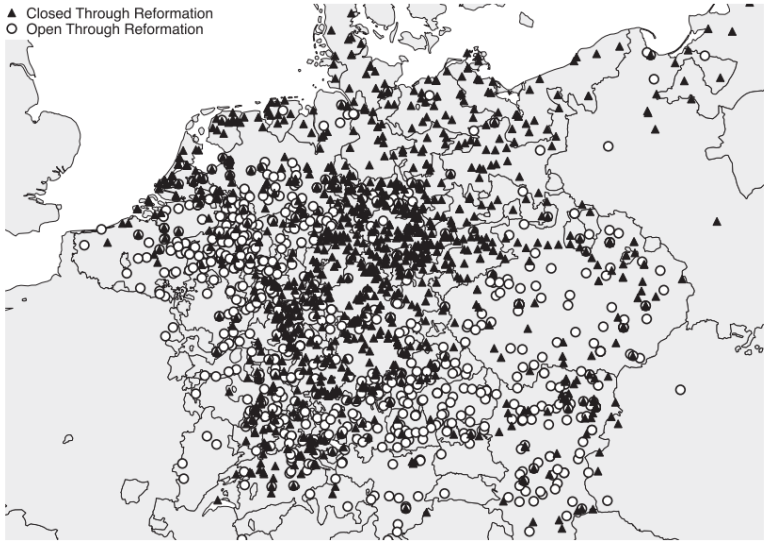
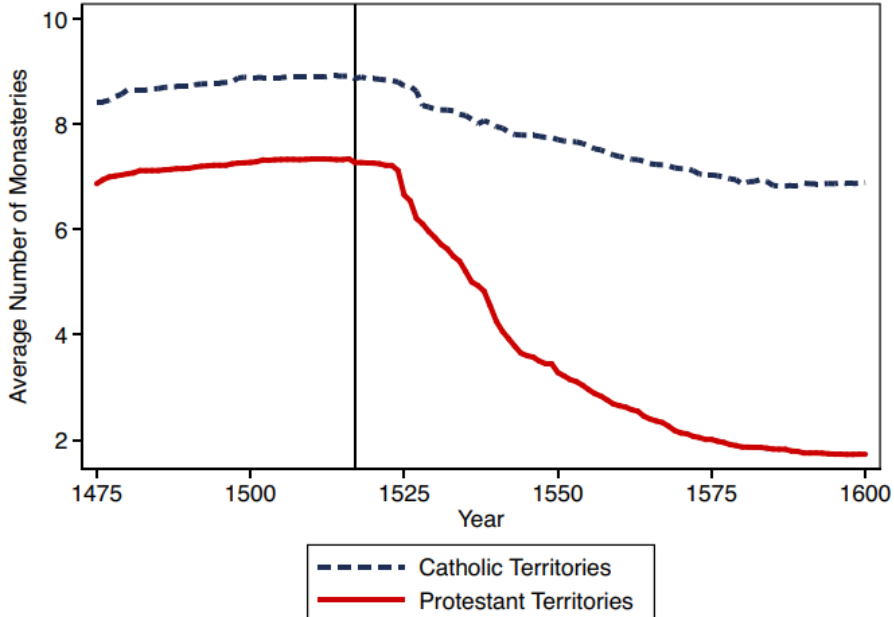


FIGURE II

Monasteries and Monastery Closure during the Reformation in Germany


Average Number of Monasteries
10
8
6
4
2
1475
1500
1525
1550
1575
1600
Year
Catholic Territories
Protestant Territories

Reading the closure figure

- Before 1517 the two lines are flat and parallel. No sign of anything coming.
- Catholic towns start with somewhat more monasteries nearby: they sit in the older, southern and western parts of the Empire.
- After 1517 density falls everywhere, and collapses where the lord went Protestant: a reduction of over **two-thirds** by 1600.
- The question the rest of the paper answers: did those resources leave the religious sector, or just move inside it?

TABLE II

MONASTERY CLOSURE AND EXPROPRIATION IN PROTESTANT TERRITORIES

Territory	Year Reformed	% Monasteries Closed	Transfer to Lord	Evidence from Wolgast (2014)
Anhalt	1537	83	Yes	Monasteries expropriated, e.g., Ballenstedt and Mehringen (76)
Baden	1555	32	No	—
Brandenburg	1539	74	Yes	Treasures taken to Berlin, covering 300,000 fl. state debt (71–72, 150)
Brunswick-Calenberg	1584	46	Yes	Expropriation and taxation to pay for ruler's servants and horses (161)
Brunswick-Lüneburg	1529	80	Yes	Expropriation of monasteries, churches (57)
Brunswick-Wolfenbüttel	1568	55	Yes	Expropriation of monasteries (230)
Denmark	1536	64	Yes	Expropriation, monastery at Husum converted to palace (see note)
East Frisia	1535	92	Yes	Conversion of monasteries to palaces and stone quarries (114–115)
Hesse	1526	76	Yes	Diversion of monastic incomes (47)
Mecklenburg	1549	75	Yes	Monastery expropriation, conversion to palace at Güstrow (195)
Nassau	1542	42	Yes	New taxes on monasteries and palace construction (see note)
Palatinate	1546	60	Yes	Expropriation of monasteries, income of 100,000 fl./year (177, 220)
Pomerania	1534	94	Yes	Monastery expropriation (101)
Ruppin	1539	56	Yes	Ruler doubles land ownership appropriating monastery lands (see note)
Saxony (ducal)	1539	96	Yes	Property seizure and sale revenues of 150,000 fl. (141–142)
Saxony (electorate)	1527	92	No	—
Upper Palatinate	1546	45	Yes	Expropriation of monasteries, income of 50,000 fl./year (220)
Württemberg	1534	67	Yes	Expropriations, silver melted down, gold seized (107)

Who ended up with the wealth

- The theologians' opening price was high: pastors, plus a "common chest" for orphans, the poor, and former monks. Wuŕttemberg monks were offered pensions of 25 guilders a year, 254 days of skilled wages.
- The price then fell. In Hesse, Landgrave Philipp drew 16,500 guilders a year from monastery lands in 1532 and 25,000 in 1565, about one-seventh of total state revenues.
- **40 percent** of monastic wealth in Hesse went to the ruler, not to religious, educational or welfare uses.
- Duke Ulrich extracted 40,000 guilders of income in Wuŕttemberg. Brandenburg's monasteries kept their privileges for a payment of 300,000 guilders.
- The lord came out richer in **16 of 18** Protestant territories.

Power moved with the money

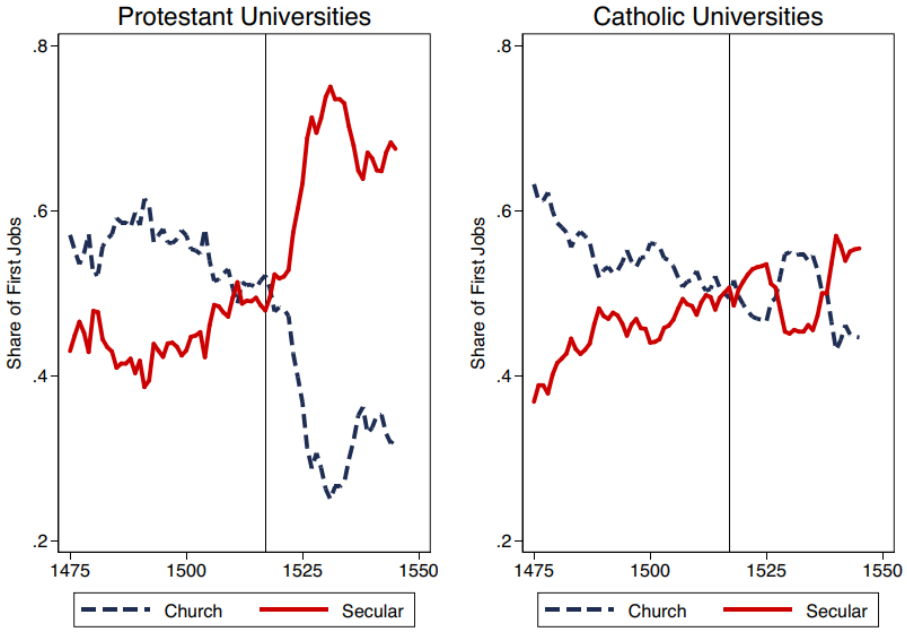
- Cohn: "Functions that had once been dispersed among several authorities became concentrated in the state. Officialdom penetrated more pervasively than before in towns, villages and families. Ecclesiastical jurisdiction was swept away."
- Lords closed monasteries rather than repurposing them, which made the shift **hard to reverse**.
- People did not stop believing. The state took over what the Church used to do and own. That is what the authors call secularization.

Three predictions to test

- Richer and more powerful lords demand more skilled labor; a Protestant church needs much less clergy.
- **Careers**: graduates move out of church jobs, and specifically into administration.
- **Degrees**: less theology, the one investment specific to a church career; more law and arts.
- **Building**: from churches toward palaces and administrative buildings.
- All three should be sharper where the Reformation won.

Data: universities

- The *Repertorium Academicum Germanicum*: the universe of recipients of German university degrees through 1550, individual by individual.
- Degrees from 1540 to 1600 hand-collected from the published registers of each university.
- Over 400 occupational titles, sorted with the *Thesaurus Professionum* into church (priests, monks) and secular (professors, judges, mayors, city councillors), with administration split out.
- **Ten universities** adopted Protestantism in a wave between 1520 and 1550, among them Wittenberg, Leipzig, Basel, Marburg, Rostock and Tübingen.
- Erfurt turned Lutheran in 1521 and back in the 1530s, so it is treated as Catholic.


Protestant Universities
Catholic Universities
Share of First Jobs
.2
.4
.6
.8
1475
1500
1525
1550
Church
Secular

Result 1: careers

- Before 1517 both groups converge on roughly half church, half secular first jobs, with no differential trend.
- After 1517 the Protestant panel flips and stays flipped. The Catholic panel keeps drifting gently.
- First job in the church sector, Protestant graduates: down about **11.8 points**, against a pre-Reformation mean of around 57 percent.
- First job in administration: up about 5 points, on a base of only about 10 percent before the Reformation.

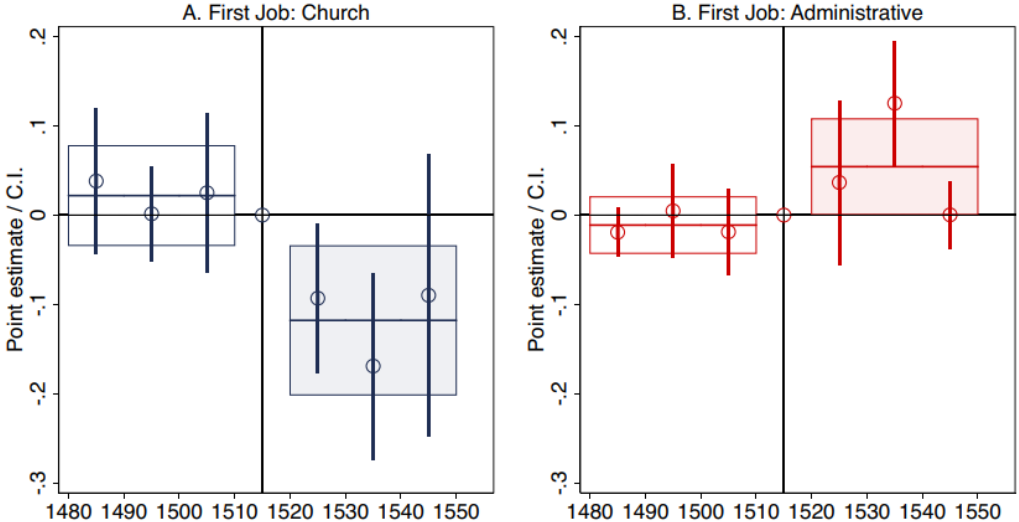


FIGURE V

Effects of the Reformation on Careers (Difference-in-Differences Estimates)

Reading the before-and-after picture

- Each dot is the gap between Protestant and Catholic universities in one decade, measured against the 1510s.
- Vertical bars show how much room for doubt there is around each dot; boxes are the averages before and after 1520.
- Before the Reformation the dots sit on zero: the two groups were on the same path.
- The break comes at the Reformation and not before it. That is why we trust the comparison.

Theology as church-specific capital

- Among graduates with at least one theology degree, **90 percent** held a church job at some point. Among those without, 53 percent.
- A theology degree is worth nothing outside the Church.
- Law and arts degrees point the other way: 27 percent take an administrative job, against 11 percent without such a degree, more than twice as likely.
- Prediction for forward-looking students after 1517: less theology, more law and arts, in Protestant universities.

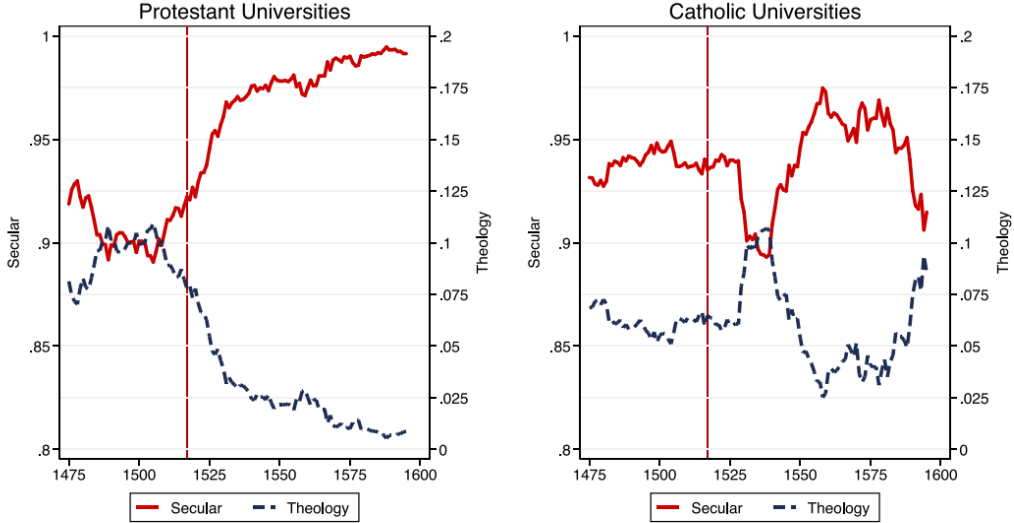


FIGURE VI

Degrees Granted in Eventually Protestant and Catholic Universities

Result 2: degrees

- Before the Reformation about 10 percent of degrees were in theology, if anything slightly more in the universities that would go Protestant.
- Afterwards theology falls **below 2 percent** of degrees granted in Protestant universities.
- Catholic universities dip mid-century, then return to the old level by 1600, alongside the Counter-Reformation.
- The gap: about 4.5 points less theology, on a pre-Reformation mean of about 6 percent, and about 2 points more law and arts.

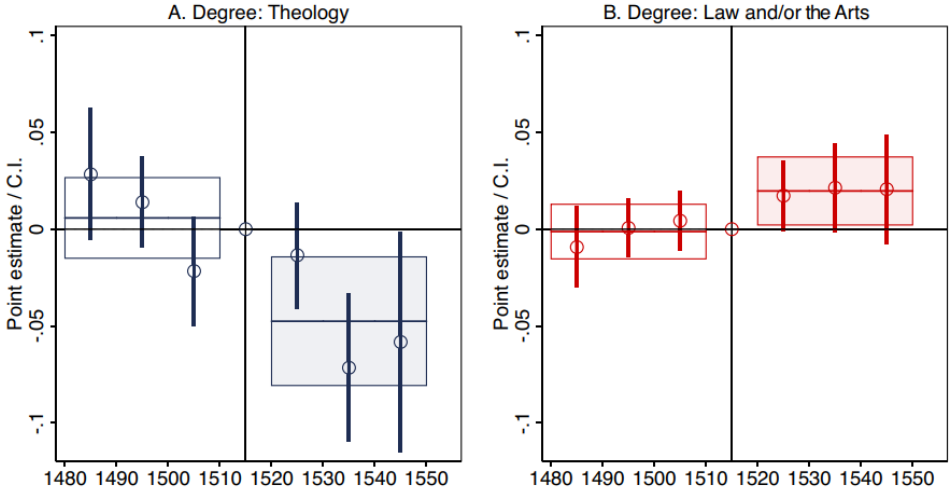
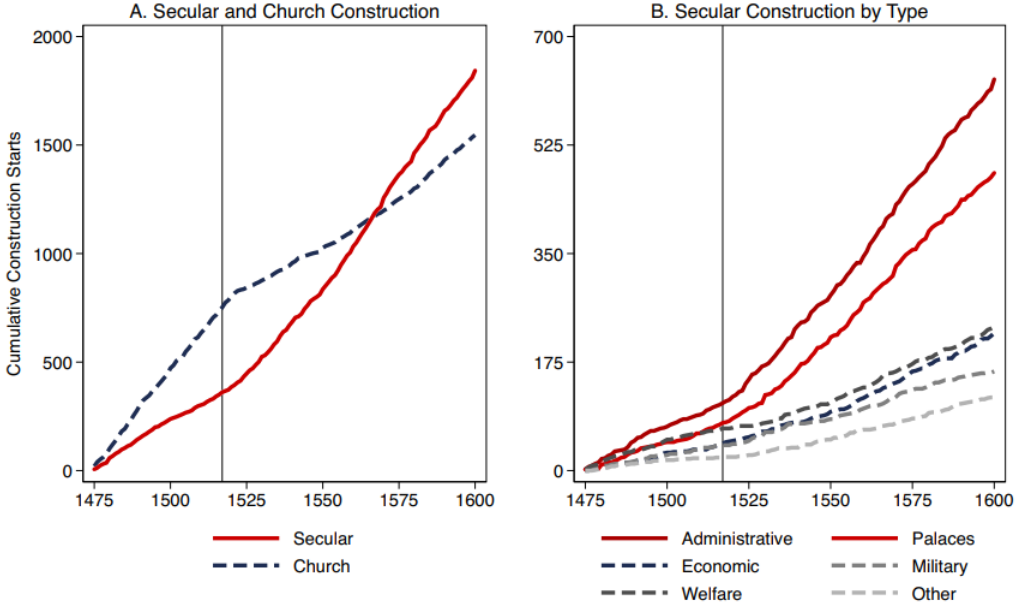


FIGURE VII

Effects of the Reformation on University Degrees (Difference-in-Differences Estimates)

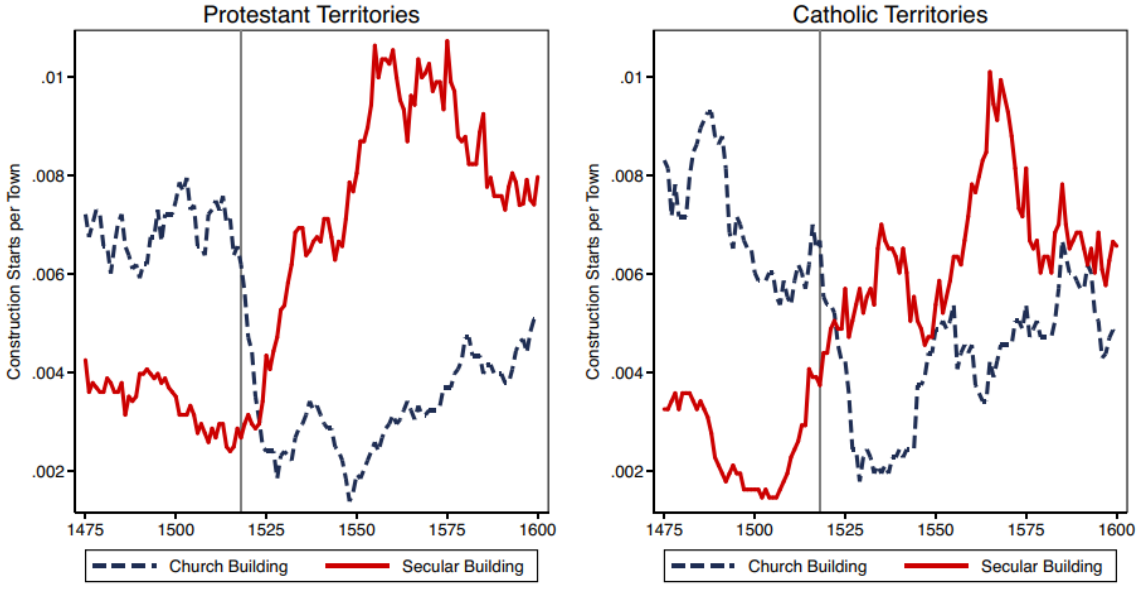
Data: construction

- Thousands of major building projects hand-coded from the "Markante Gebaüde" entries of **2,256 towns** in the *Deutsches Städtebuch*.
- Each recorded by place, start year and sector: church (churches, monasteries) or secular.
- Secular split further: administrative (courts, town halls, customs houses), palaces (castles, hunting lodges), welfare (schools, hospitals), economic (mills, breweries), military (barracks, arsenals).
- A construction start bundles land, finance, labor and materials, so it summarizes where resources went.
- Rare events: a German town had one to two major construction starts per century.


A. Secular and Church Construction
Cumulative Construction Starts
2000
1500
1000
500
0
1475
1500
1525
1550
1575
1600
Secular
Church
B. Secular Construction by Type
700
525
350
175
0
1475
1500
1525
1550
1575
1600
Administrative
Economic
Welfare
Palaces
Military
Other

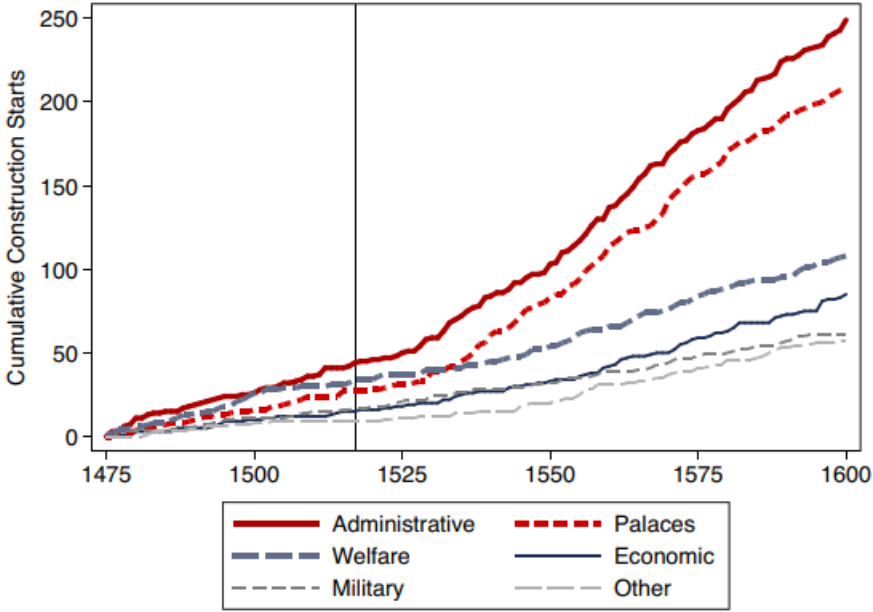
Germany as a whole

- Left panel: cumulative church and secular starts. Church building is ahead in 1517 and behind by the end of the century.
- The pivot happens **at** the Reformation, not before it.
- Right panel: inside the secular sector the pull-away categories are administrative buildings and palaces.
- Welfare, economic and military building grow far more slowly.


Protestant Territories
Catholic Territories
Construction Starts per Town
.002
.004
.006
.008
.01
1475
1500
1525
1550
1575
1600
Church Building
Secular Building

Result 3: building, by religion of the lord

- Protestant territories: church building falls, secular building rises sharply, and stays above church building for the rest of the century.
- Catholic territories: a milder and less durable version of the same movement.
- The increase concentrates **after 1550**, once title to the expropriated property was settled.
- 1550 to 1599: about 1.5 extra secular starts and 0.75 extra palaces or administrative buildings per territory per decade, against pre-Reformation means of 2.2 and 1.4.


Cumulative Construction Starts
250
200
150
100
50
0
1475
1500
1525
1550
1575
1600
Administrative
Palaces
Welfare
Economic
Military
Other

What it is not

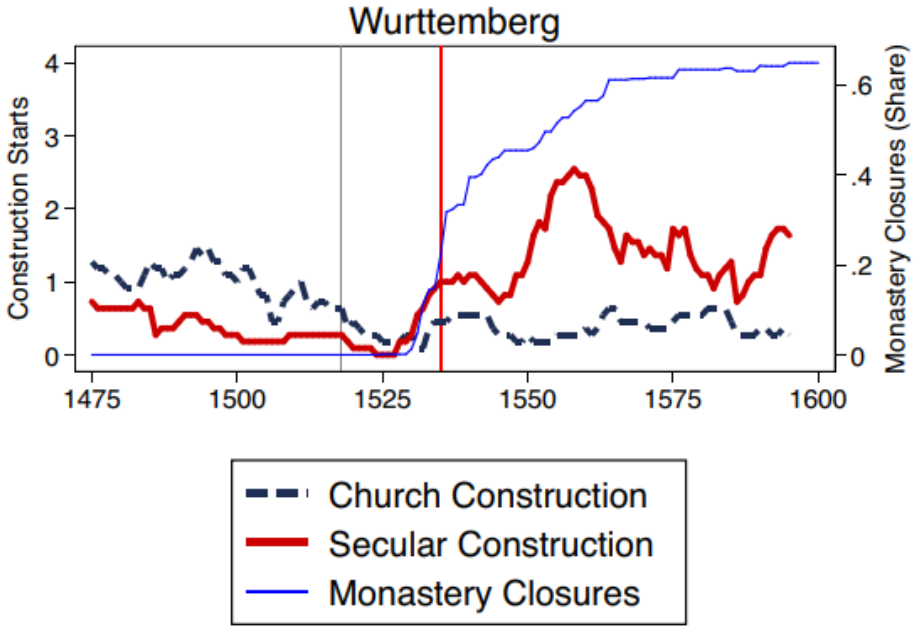
- Not relabeling: welfare building, schools and hospitals, the genuinely ambiguous category, does not shift.
- Not religious warfare: no rise in military building through 1600.
- What rises is **palaces and administrative buildings**, the uses that reflect the wealth and power of secular lords.
- Competition intensity: the gap comes from comparing Protestant regions with Catholic regions far from a Protestant border. Catholic towns on the border look Protestant.

Could something else explain the timing?

- No differential pre-1517 trends in careers, degrees or building between the two groups.
- Cities led the Reformation, so perhaps this is just urban change. But the same pattern appears in the small towns.
- Remaining worry: an economic shock that caused both the conversion and the reallocation.
- Answer: three territories where the **date** of religious change turned on an unexpected succession.

Three successions

- **Brandenburg**: Joachim I, who had bought an archbishopric for his brother with loans against future indulgence sales, made his son sign a contract promising to stay Catholic. He died in 1535 and Joachim II reneged at once.
- **Ducal Saxony**: Duke Georg's last Catholic son died in 1539; he began transferring the inheritance to the Catholic Ferdinand but died before it was complete, and the Protestant Heinrich took over.
- **Württemberg**: Ulrich was exiled in 1519 after killing his mistress's husband, adopted Protestantism in 1523, and recovered the duchy in 1534 with Philip of Hesse's help. By 1535 one-third of the duchy's farmland had passed from the monasteries to him.


Wurttemberg
Construction Starts
0
1
2
3
4
Monastery Closures (Share)
0
.2
.4
.6
1475
1500
1525
1550
1575
1600
Church Construction
Secular Construction
Monastery Closures

Summing up Cantoni, Dittmar and Yuchtman

- Introducing religious competition moved resources and political power from religious elites to secular lords.
- They call it **economic secularization**, in two parts: institutional (stronger states) and allocational (labor, education and building).
- This runs through princes and property, nothing to do with Weber's work ethic.
- First quantitative evidence that the Reformation played a causal role in the secularization of the West.

An aside: the Iberian world

- A question the papers do not ask: Portugal, Spain and their colonies never had a rival church.
- The Church there kept the monopoly and the land. Brazil's church, orders and missions accumulated estates for centuries.
- Expropriation still happened, but as a purely secular act with no theology attached: the Jesuits expelled from the Portuguese empire in the eighteenth century, their estates taken by the crown.
- So did the princes need Luther, or just an army?

The two papers together

- **Iyigun**: why it was not crushed in its infancy, decided partly on the Hungarian frontier.
- **Cantoni, Dittmar and Yuchtman**: the Church's land and its graduates passed to the princes.
- Next class: the same expropriation in England, and what it did to the economy there.